

Assessing the Robustness of TerraMind to Sensor Domain Gap for On-board Processing on the Φ -sat-2 Mission

MASTER'S THESIS

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Statement on the Usage of Artificial Intelligence

AI Tool	Purpose / Use Case	Section of Thesis
Claude Code (Academic Research Skills)	Support for methodology framing	Chapter 3
Claude Opus 5	Detailed below*	Chapter 1 to Chapter 5
Gemini 3.1 Pro	Tikz Code Generation	Chapter 3, Figure 3.16

Table 1: Usage of Artificial Intelligence Tools

* Claude Opus 5 (Anthropic) was used during the preparation of this thesis, for the review of chapters, for critique of the argument structure, and for the formatting of paragraphs in $\text{\LaTeX}$.

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List of Acronyms

API	Application Programming Interface
BN	Batch Normalization
CNN	Convolutional Neural Network
DA	Domain Adaptation
DEM	Digital Elevation Model
DN	Digital Number
DSBN	Domain-Specific Batch Normalization
EO	Earth Observation
ESA	European Space Agency
GeoFM	Geospatial Foundation Model
GRD	Ground Range Detected
HDF5	Hierarchical Data Format, version 5
IoU	Intersection over Union
KD	Knowledge Distillation
KL	Kullback–Leibler (divergence)
LULC	Land Use / Land Cover
mIoU	mean Intersection over Union
MMD	Maximum Mean Discrepancy
MTF	Modulation Transfer Function
NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared
NRT	Near Real-Time
ONNX	Open Neural Network Exchange

Chapter 1

Introduction

Earth observation (EO) satellites now capture more data than ground segments can downlink and process fast enough[5]. Ground-based processing is enough for most applications, but the time-critical ones such as flood mapping[6] and wildfire detection[7] need near real-time (NRT) inference, since the events they monitor can evolve within hours. Moving the inference step from the ground to the satellite itself removes this constraint: instead of downlinking large raw imagery for later processing, an on-board model directly downlinks a small processed data product. A growing number of missions have begun to demonstrate that this is technically feasible[5]. The European Space Agency’s Φ -sat-2 mission is among the most complete demonstrations to date: a 6U CubeSat carrying a high-resolution multispectral imager alongside a space-qualified Intel Movidius Myriad 2 VPU (Vision Processing Unit)[8]. Instead of running a single fixed application, Φ -sat-2 uses the NanoSat MO Framework[9] to install, update and orchestrate several AI applications on board, much as a smartphone runs installable apps.

The applications created so far to run on board Φ -sat-2 are small, task-specific convolutional neural networks (CNNs), each one trained, validated and uplinked individually for a specific job such as cloud masking, vessel detection or roadmap extraction[10]. Adding a new capability to the system therefore means repeating the entire pipeline from scratch. Geospatial Foundation Models (GeoFMs) offer a different model of development. Pre-trained on large and diverse archives of EO imagery, a GeoFM such as TerraMind[11] learns a general-purpose semantic representation that downstream tasks can exploit with comparatively few labeled examples, and without a full training run for every application. This advantage on few-shot learning is useful for emergency applications such as flood and burnt-area mapping because labeled events are rare and new missions don’t have time to accumulate annotated events.

Two challenges separate this promise from an operational on-board GeoFM. The first one is architectural, and it comes from the limits of the target hardware. TerraMind is built on the Vision Transformer (ViT)[12] architecture, which is structurally incompatible with the space-qualified accelerators of the Myriad 2’s generation[8], since that hardware predates the support ViT inference requires. The second one is the operational domain gap, and it comes from the sensor differences. GeoFMs such as TerraMind are pre-trained on Sentinel-2 and similar archived missions, but Φ -sat-2 has a different sensor, and the properties of its imagery are different enough from Sentinel-2’s that the model might not generalize well to it. This phenomenon is known as a domain gap[13]. Neither problem solves the other: a compressed model built without addressing the domain gap simply compresses the gap with it, and a domain-adapted ViT still cannot run on the target

a limited set of real target-domain labels, with the encoder kept frozen. However, the source and target domains they compare come from different benchmark datasets and not from co-registered pairs, so the sensor-induced shift is not isolated from the content and geographic shift.

Within the Φ -lab, a **GeoFM** named Φ satNet was developed specifically for on-board execution on Φ -sat-2 hardware. Built on the U-Net architecture (**CNN**-based), it avoids the **ViT**-compatibility problem, and it is pre-trained from scratch on physics-simulated Φ -sat-2 data. Neither of these works isolates the sensor-only covariate shift using co-registered data, and neither tests whether an existing **GeoFM**'s broad and diverse pre-training transfers to a new sensor with fewer labels. This thesis addresses the first point directly; the second one involves a matched label-efficiency comparison against Φ satNet. The gap this thesis addresses is therefore threefold: (i) whether the domain gap between Φ -sat-2 and Sentinel-2-pretrained TerraMind can be measured directly, in latent space, using data that isolates the sensor-induced covariate shift; (ii) whether that gap can be closed through an explicit domain-adaptation mechanism; and (iii) whether the resulting adaptivity survives compression into a hardware-deployable, cross-architecture (**ViT**-to-**CNN**) student model.

1.3 Research Objective and Problem Statement

The objective of this thesis is to determine whether the representational benefits of a large-scale, Sentinel-2-pre-trained **GeoFM** can be carried across the sensor domain gap into a lightweight **CNN** that has to operate on Φ -sat-2 imagery and to run on the Φ -sat-2 on-board processor. TerraMind is not itself adapted to Φ -sat-2, so we use it as a frozen teacher on the Sentinel-2 domain it was pre-trained for. To bridge the gap, the student is trained on both domains, so that it learns to represent the two sensors while being deployed on only one of them. The question is therefore whether a student distilled under this arrangement represents the two sensors consistently and performs well on the target sensor, without giving up the advantage that motivated starting from a **GeoFM** instead of training from scratch in the first place. This objective is pursued through three research questions:

1. Does the domain gap caused by the sensor differences between Φ -sat-2 and Sentinel-2 measurably affect the latent representations produced by TerraMind, and if so, where in the encoder does this effect concentrate? And since the two instruments differ in several respects at once, which of those physical differences is the one that produces the gap?
2. Can a lightweight **CNN** distilled from TerraMind be made sensor-adaptive, given that a naive fine-tuning on degraded data alone does not achieve it? And if it can, which mechanism is responsible: the domain-specific normalization, a paired-contrastive objective exploiting the co-registered triplets, or both?
3. Does distilling from TerraMind in this way improve the downstream Φ -sat-2 task performance over simply training a compact model of the same architecture directly on the target sensor? Sensor-adaptivity and task accuracy are assessed as two separate questions here, without assuming that the first one implies the second.

Chapter 2

Background and Literature Review

This chapter positions the thesis against the strands of work that each contribute to the overall problem, which is to make the representational benefits of a **GeoFM** available on board a resource-constrained satellite. We present them in turn, and a closing synthesis states what combination of them remains unaddressed.

2.1 On-board AI for Earth Observation

The volume of data returned by Earth observation missions keeps growing, driven both by the number of platforms in orbit and by the increasing capability of their instruments, whether through a finer spatial resolution or, in the hyperspectral case, a far denser spectral sampling. The result is a body of products that covers the globe repeatedly and serves a wide range of applications. Extracting value from it means moving from observations to information, and ultimately to decisions taken by users who are not themselves remote sensing specialists. Machine learning is well suited to automating that step, since it can interpret the complex, high-dimensional structure of **EO** acquisitions for tasks such as predictive analytics and anomaly detection [5].

The conventional way of applying such models keeps the whole inference stage on the ground. Models are trained on datasets compiled from the archives of one or several missions, and each new acquisition is downlinked in full before any information is extracted from it. This is the bent-pipe architecture (Figure 2.1), in which the satellite acts as a passive relay: it acquires the data, applies at most a minimal preprocessing, and transmits the result to a ground station without interpreting it [1]. The arrangement keeps the mission design simple, but it also makes the useful output of a mission dependent on ground infrastructure, and bounds it by downlink bandwidth and latency, which is exactly what prevents a near-real-time response.

Because the satellite is not part of the information extraction process under this arrangement, and because modern missions add to an already large archive faster than it can be exploited, there is a growing interest in running the inference on board, on the imagery as the sensor captures it. Not every platform can host such a model. Image-processing networks, and convolutional and deep neural networks in particular, are poorly matched to the processors typically flown, which are chosen for their reliability and their power efficiency instead of their throughput, and which offer a limited computation and memory [5]. An AI-enabled satellite therefore carries a model that is uplinked from the ground and trained for that mission specifically.

Φ -sat-1 was the first mission to demonstrate that this is achievable, by running a

dynamic-axes support renders transformer-based architectures such as ViT incompatible with the target [21]. These constraints are the main motivation for a cross-architecture distillation, in place of a direct deployment or a same-family compression.

2.2 Geospatial Foundation Models

Foundation models are a class of large, pre-trained neural networks that can be adapted to many downstream tasks with a minimal additional training [22]. They are pre-trained with self-supervised learning on massive and diverse datasets, which is what gives them the ability to generalize across tasks and domains. The approach is now widely adopted in natural language processing and in computer vision, where models such as BERT, GPT and CLIP perform well across a range of tasks. Once it is pre-trained, such a model is fine-tuned per task instead of being trained from scratch.

GeoFMs [23] extend this paradigm from natural images to Earth observation. Pre-training a single large backbone on diverse, multi-modal EO archives means that many downstream tasks can be solved by fine-tuning a small task-specific head, instead of training a full model from scratch. TerraMind [11] is a ViT-based, multimodal GeoFM pre-trained on the multimodal TerraMesh dataset [24], which includes over 9 million multi-seasonal samples with eight spatiotemporal aligned modalities (Sentinel-2 at L2A, L1C, RGB, Sentinel-1 RTC and GRD, DEM, NDVI, LULC). It is explicitly designed to produce transferable semantic representations rather than task-specific features. However, that capacity has so far mostly been demonstrated on data drawn from the same sensor family the model was pre-trained on, so how it behaves under different sensor conditions is still an open question.

PANGAEA [25] puts the generalization claim behind GeoFMs to a systematic test. It was created because the existing GeoFM evaluations were narrow and inconsistent: the benchmarks were often geographically biased toward North America and Europe, and several of them assessed the models through a single instrument. PANGAEA spans eleven datasets across urban, agricultural, marine and forest domains, with a global coverage, spatial resolutions from 1.5 to 30 m per pixel, uni- to multi-temporal configurations, and several sensor types including multispectral, natural-colour and synthetic aperture radar imagery. The models are compared against supervised task-specific baselines, among them a plain U-Net and a vanilla ViT.

The finding that matters the most for this thesis is that GeoFMs do not consistently outperform those supervised baselines, and that task-specific models exploit the spectral and spatial detail of a dataset more effectively when the characteristics of the pre-training data differ from those of the downstream data. The evaluation identifies the spectral and spatial resolution of the pre-training sensor as a significant determinant of the downstream performance, especially when the downstream task requires fine-grained spatial or spectral information. The authors also observe that geospatial models are evaluated mainly on narrow selections of tasks and datasets, which leaves their ability to handle domain shifts across geographic regions, sensors and temporal conditions largely unexplored.

PANGAEA does compare across sensors, in the sense that its datasets originate from different instruments, but the sensor is never the only variable that changes between them. Separating that factor from the others requires observations of the same ground by two instruments, which is what the dataset described in Chapter 3 supplies, and what the measurements of Chapter 4 rest on.

cloud detection in the context of the Φ -sat-1 mission. They proposed novel task definitions for **DA** which they named *offline adaptation* and *online adaptation*. Offline adaptation relies on **SDA** and downlinking to update the **CNN** remotely as new labeled data of the target domain becomes available during the mission. Online adaptation demonstrates *test-time* adaptation, where the model is updated on-board with unlabeled data, using a self-supervised loss.

This thesis focuses on an initial **SDA** step, where a small amount of labeled target data is available for adaptation and evaluation, and does not address the online adaptation problem. The main author also reports a follow-up work [15], where they deployed a compressed Prithvi-family **GeoFM** on Kanyini and the IMAGIN-e payload, and adapted it via a real-label head retraining instead of an encoder-level **DA**. They measure the domain gap only indirectly, through the downstream task performance on source and target benchmark datasets that differ in geography and content as well as in sensor. This is the central limitation that the data design of this thesis addresses, because without co-registered source/target pairs of the same scene, a measured performance drop cannot distinguish the sensor-induced covariate shift from the confounding differences in what the two datasets depict.

Beyond retraining task heads, the approaches to close the gap fall into a small number of families. What separates them is how each one attacks the divergence term of the bound introduced at the start of this section. Discrepancy-based methods penalize a statistical distance between the source and target feature distributions directly, most commonly the maximum mean discrepancy (**MMD**) [28], which requires only that samples from each domain are available. Alignment-based methods make the stronger assumption that corresponding inputs are available, and pull their representations together explicitly. X-STARS [29] is a close example to the setting of this thesis. It adapts a model pre-trained on one optical sensor to a second sensor through a multi-sensor alignment dense loss, which is a patch-wise contrastive objective applied between paired acquisitions of the same area by both sensors, within a continual-pretraining framework. It establishes that a paired-contrastive objective is effective when the sensor change is primarily one of spatial resolution, which is not the regime this thesis operates in: the Φ -sat-2/Sentinel-2 gap measured in Chapter 4 is driven mainly by a differing spectral response.

A separate line of domain-adaptation work avoids the alignment losses entirely, by giving the model domain-specific normalization statistics instead. Domain-Specific Batch Normalization (**DSBN**) [4] is a generic technique, originally for **UDA**, and it is designed to be integrated into deep networks that use batch normalization layers. Batch normalization (**BN**) [30] is a technique used in deep learning to normalize the inputs of each layer, which helps stabilize and accelerate the training. **DSBN** extends **BN** by keeping a single set of convolutional weights, but a separate batch-normalization affine transform and separate running statistics (mean and variance) per domain. This only requires that the domain of each example is known, which is a much smaller requirement than the paired-contrastive or **MMD** loss objectives discussed above, and, unlike them, it is directly applicable wherever the deployed sensor is known at inference.

2.4 Physics-Based Sensor Degradation Simulation

Because labeled data from a satellite sensor does not exist before the satellite is launched, and accumulates slowly afterward for rare events, a physics-based simulation of a target

co-registered triplets was implemented and evaluated as an additional mechanism on top of this, but it was found not to improve the downstream task performance beyond what the domain-specific normalization already provides (Section 3.4).

2.6 Φ satNet

Φ satNet is a U-Net-derived, multi-task-pretrained GeoFM being developed by the Φ -lab specifically for on-board execution on Φ -sat-2, using a purpose-built pre-training corpus and a benchmark suite built on the same OrbitalAI simulator described above. By training a compact, non-transformer architecture from scratch, Φ satNet avoids the ViT hardware-incompatibility problem entirely, and its authors report that it matches or exceeds unconstrained transformer-based GeoFMs on several on-board tasks [21]. This is a different design philosophy from the one pursued here. Φ satNet optimizes for a hardware-native architecture trained on a curated, mission-specific task suite, while this thesis asks whether an existing, diversely pre-trained GeoFM can be adapted to new tasks with fewer labels than training a mission-specific model requires.

One property of Φ satNet is directly related to the problem this thesis addresses. Its pre-training method and its n -shot benchmarks are built entirely on simulated Φ -sat-2 imagery derived from Sentinel-2, and not on real acquisitions [21]. Φ satNet therefore does not solve the sensor domain gap by being mission-specific. Its reported few-shot advantage rests on the assumption that a physically simulated imagery stands in for the real sensor, and its base being simulated, cannot test that assumption. A comparison against Φ satNet on a shared task and label budget is the appropriate test of the label-efficiency claim.

2.7 Synthesis

Through all of the works cited above, none isolate sensor-induced covariate shift using co-registered source/target data. This thesis gives a cross-architecture distillation student an explicit, domain-aware adaptation mechanism, and verifies that the task accuracy transfers into hardware-deployable form. The spatially and temporally co-registered Φ -sat-2/Sentinel-2/simulated triplet dataset used in this thesis is what makes that isolation possible; the domain-specific batch normalization applied during cross-architecture distillation (Chapter 3) is what makes the adaptation and its verification possible. The co-registered triplets also serve to show that a paired-contrastive alternative does not do better than the simpler domain-specific normalization. The real-acquisition component also separates this work from the mission-specific alternative as Φ satNet is pre-trained and benchmarked on simulated Φ -sat-2 imagery, so the assumption that simulation stands in for the real sensor is one both approaches make and only real, co-registered data can check.

Co-registered views, independently contrast-stretched

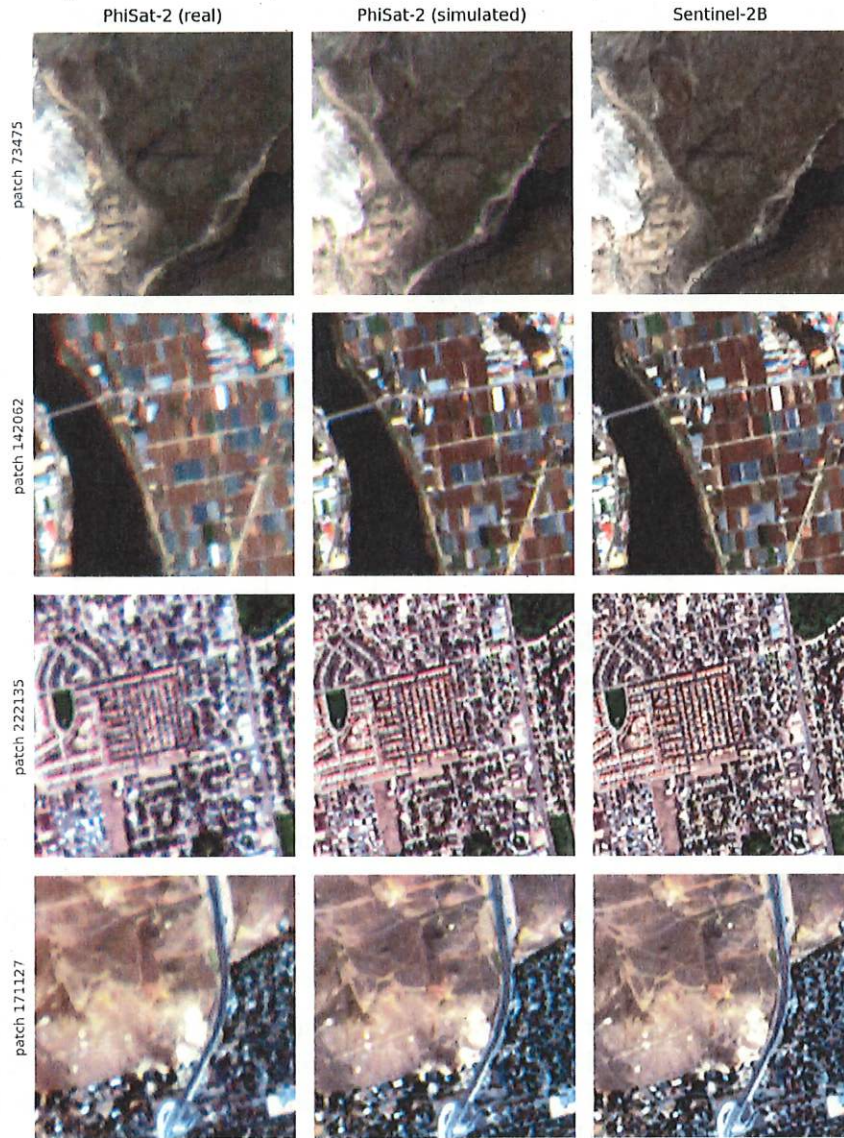


Figure 3.1: Four representative triplets, each row showing the same scene as observed by real Φ -sat-2, the physics-simulated Φ -sat-2 rendering, and Sentinel-2B (each channel independently contrast-stretched for display). The real acquisitions shows blurrier scenes despite the higher resolution and a noisy texture compared to the simulated or source Sentinel-2 view.

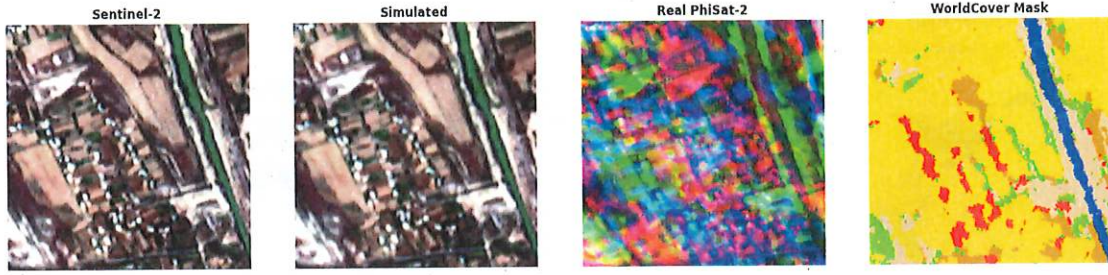


Figure 3.3: A representative triplet, each row showing the same scene as observed by real Φ -sat-2, the physics-simulated Φ -sat-2 rendering, and Sentinel-2B (each channel independently contrast-stretched for display). The real acquisitions shows heavy band misalignment artifacts compared to the simulated or source Sentinel-2 view.

Scale and coverage

After discarding a small number of corrupted source products, the dataset comprises 253,228 co-registered triplet patches drawn from 1,299 distinct Φ -sat-2 products (individual acquisitions), a median of 209 patches per product (up to 713 for the largest).

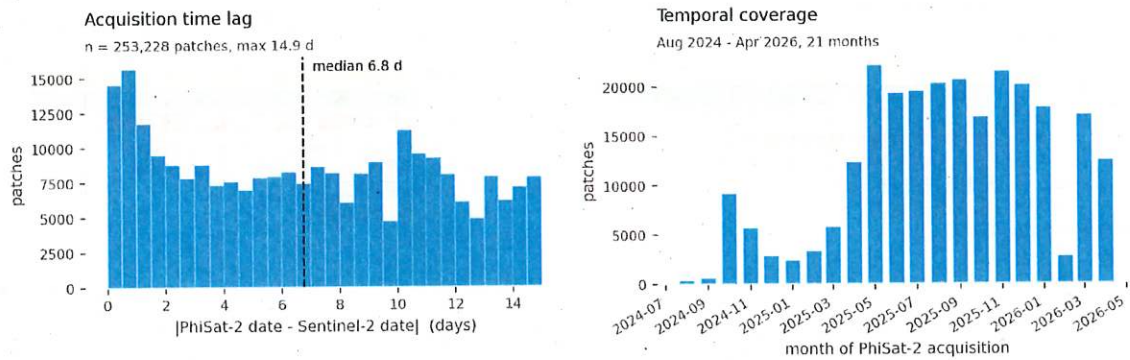


Figure 3.4: Left: distribution of the absolute time lag between matched Φ -sat-2 and Sentinel-2 acquisitions (median 6.8 days, maximum 14.9 days, $n = 253,228$). Right: monthly acquisition volume, Aug. 2024–Apr. 2026, showing the collection ramp-up in the first eight months.

Acquisitions span 2024-08-21 to 2026-04-15 (21 months), with collection volume higher from mid-2025 onward than in the initial period. Real- Φ -sat-2 and Sentinel-2 acquisitions are not simultaneous: their dates are matched to within a median of 6.8 days and a maximum of 14.9 days, an acceptable temporal offset for a largely stable target such as land cover. 73.1% of retained patches have under 5% total cloud cover, and only 0.199% of pixels are unlabelled after the [ESA](#) WorldCover rasterization step.

Geographically, the 1,299 source products span latitudes -54° to 76° with global but uneven coverage (Figure [3.5](#)). By Köppen–Geiger climate group, the patch population is dominated by arid (41%) and temperate (29%) climates, with tropical (14%), continental (10%), and polar (1%) regions under-represented (4% unclassified).

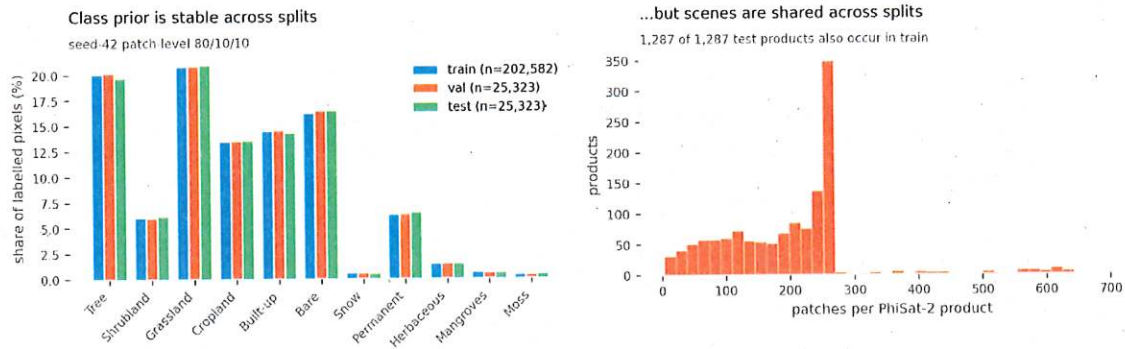


Figure 3.7: Left: class prior is stable across the train/validation/test split. Right: every one of the 1,287 products contributing to the test set also contributes patches to the training set as the split is patch-level.

Measured sensor discrepancy

Three measurements quantify the sensor discrepancy, without any model being involved.

First, the raw radiometric scales of the two sensors differ substantially (Figure 3.8): real Φ -sat-2 digital numbers are concentrated below Sentinel-2 and simulated- Φ -sat-2, whose density curves overlap so closely in every band that the Sentinel-2B trace is entirely hidden beneath the simulated one. However, the distribution of the real Φ -sat-2 is not simply shifted lower. Alongside its dominant low-value, it carries a smaller secondary lobe that reaches into the same value range as Sentinel-2/simulated, and that overlaps with their peak in every band. This secondary lobe is consistent with the measurable saturation (2.1–2.2% of pixels in the RE1/RE2 bands) and dropout (0.3–0.36% in RE2/RE3) already established for the real sensor: a small group of real pixels sits far from the sensor’s typical response. At this level, the simulated and the raw Sentinel-2 domains remain close to statistically indistinguishable, which confirms that the physics-based degradation pipeline (Section 3.2) perturbs the fine spatial and spectral structure while leaving the intensity statistics largely untouched. The domain gap this thesis is concerned with is therefore a real-vs-simulated gap, not a simulated-vs-source gap.

Per-domain normalization (Section 3.3.1) aligns the three distributions, but their shape still differ (Figure 3.9). Sentinel-2B and simulated Φ -sat-2 remain closely overlapping, while real Φ -sat-2 is distinctly bimodal in every band: a high peak near $z \approx -0.7$, and a separate peak around $z \approx +0.5$ to $+1.5$. In the red and red-edge bands (B04–B08) this second spike sharpens further into a narrow spike near $z \approx 2.3$ – 2.5 , consistent with the same RE1/RE2-concentrated saturation noted above: the smooth clip in the normalization protocol (Section 3.3.1) compresses these values instead of removing them. The domain-adaptation mechanism of Section 3.4 must close this shape mismatch.

The second measurement is the most directly relevant one to the degradation-driver hypothesis discussed in Chapter 4. The shape of the spectral signature, which is each sensor’s per-band median normalized by its own overall band mean so that the absolute scale is divided out, differs between real Φ -sat-2 and Sentinel-2/simulated- Φ -sat-2 (Figure 3.10). Sentinel-2 and the simulated domain are, again, visually indistinguishable from each other. Real Φ -sat-2, in contrast, shows a pronounced peak at the red band (1.85 relative to its own mean, against 0.85 for Sentinel-2/simulated) and a dip at the near-infrared band (0.70 against 1.21). This analysis gives us a quantitative support for attributing the domain gap primarily to a spectral response function mismatch (Section 4.2), and not

to the spatial resolution or the noise level, since a resolution-driven or noise-driven gap would not be expected to alter the relative shape of the spectral signature this distinctly.

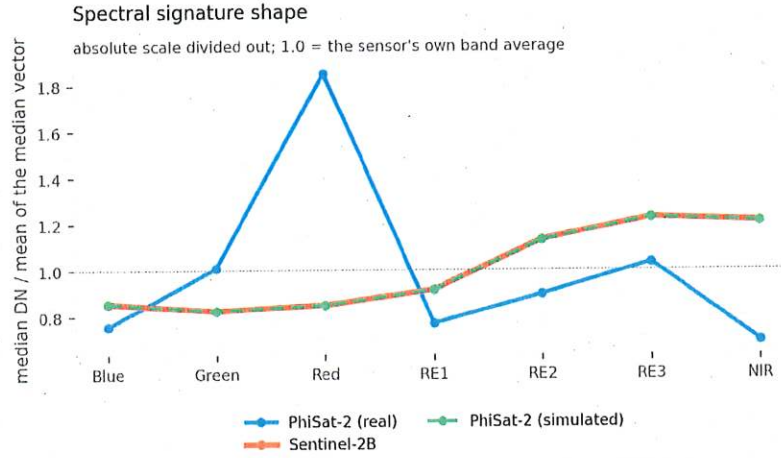


Figure 3.10: Median per-band digital number, normalized by each sensor's own overall band mean (1.0 = that sensor's average). Sentinel-2B and simulated Φ -sat-2 overlap almost exactly; real Φ -sat-2 has a markedly different spectral shape, peaking at Red and dipping at RE1 and NIR.

These measured per-domain statistics (mean, standard deviation and clip value per band) are computed separately for the real, the Sentinel-2 and the simulated domains. They parametrize the per-domain normalization protocol of Section 3.3.1.

The third measurement is that the residual disagreement remaining after per-domain normalization can be localized spatially, and broken down by land-cover class. For a pair of normalized views of the same patch, a per-pixel approximation error is defined as the mean-over-bands ℓ_1 distance,

$$\text{err}(u, v) = \frac{1}{7} \sum_{c=1}^7 |x_c^{\text{real}}(u, v) - x_c^{\text{s2b}}(u, v)|, \quad (3.1)$$

explain

where x^{real} and x^{s2b} are the per-domain-normalized (Section 3.3.1) real- Φ -sat-2 and Sentinel-2B views of the same co-registered patch. Because both views are already standardized on their own statistics, Equation (3.1) isolates the part of the gap that the marginal alignment does not explain: the point spread function, the band response, the view geometry and the residual co-registration error, rather than the location/scale offset that the normalization protocol already removes.

Aggregated over 5,000 patches and broken down by the dominant ESA WorldCover class of each pixel, the pixel-weighted mean approximation error is 0.91 (Figure 3.11). Among the five best-supported classes (13–21% of pixels each), the error is not uniform: bare/sparse vegetation and permanent water (≈ 1.06 – 1.07) disagree between sensors noticeably more than tree cover and shrubland (≈ 0.73), with cropland, built-up, and grassland in between (≈ 0.85 – 0.88). Open, low-texture or specular surfaces (bare ground, water) are plausibly more sensitive to the residual radiometric and geometric mismatches that Equation (3.1) captures, than vegetated and texturally rich surfaces are, since the latter anchor the co-registration and the view-geometry effects more consistently across

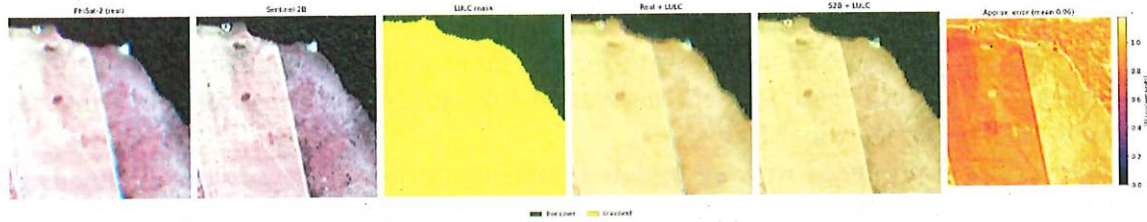


Figure 3.13: Patch showing a grassland cutting a forest. This simple scene has a low mean approximation error (0.96).

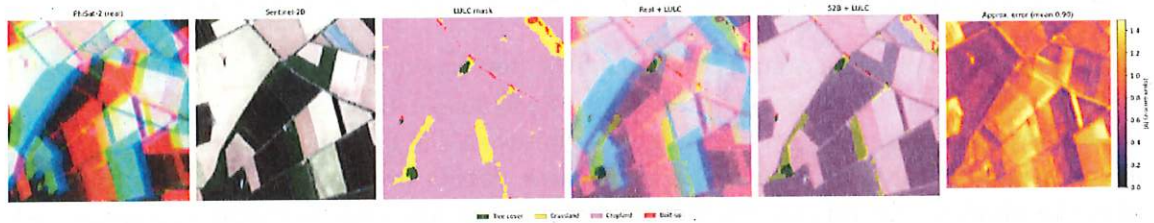


Figure 3.14: Patch showing cropland (mean approximation error 0.90). This real data shows the band misalignment that Equation (3.3) models synthetically. The scalar mean approximation error for this patch (0.90) is close to the dataset-wide mean despite the heavy artifact: Equation (3.1) averages absolute per-band intensity differences and is insensitive to spatial/geometric coherence, so it under-represents local misalignment, which is a limitation of the metric.

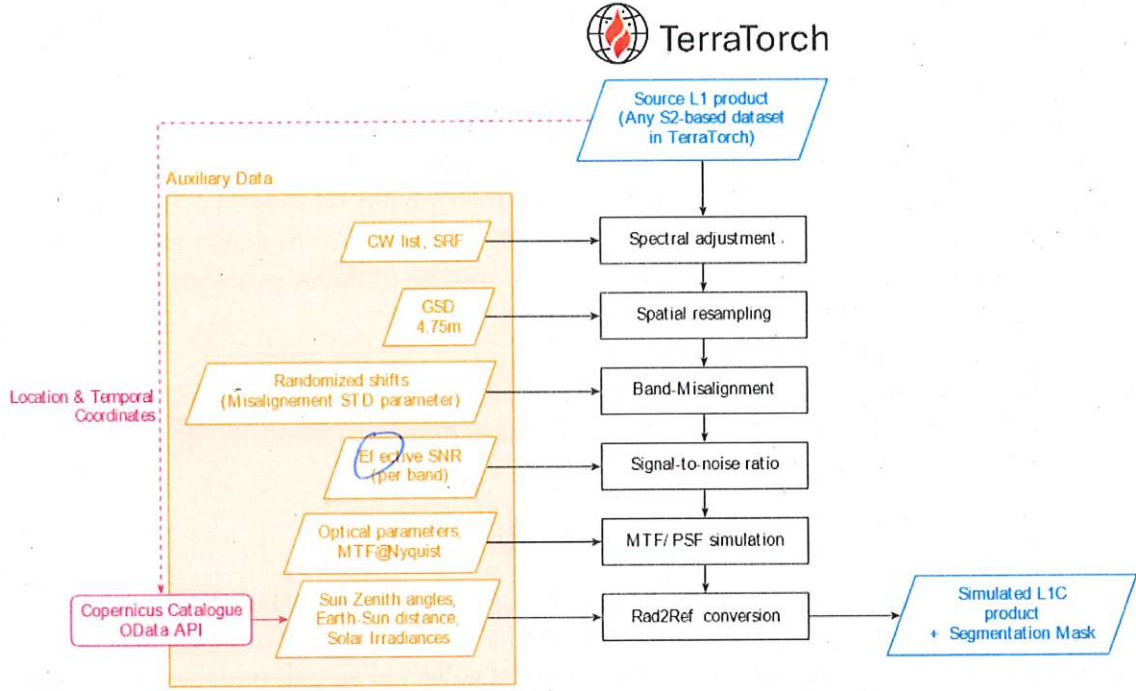


Figure 3.15: The updated degradation pipeline. Source Sentinel-2 L1 product passes through five physical stages in radiance space before conversion back to reflectance, each driven by its own auxiliary input.

Radiance–reflectance conversion. TOA reflectance ρ and TOA radiance L are related by

$$\rho(i, j, b_n) = \frac{\pi d^2}{E_{\text{SUN}}(b_n) \cos \theta_S(i, j)} L(i, j, b_n), \quad (3.2)$$

where d is the Earth–Sun distance in astronomical units, $E_{\text{SUN}}(b_n)$ the exo-atmospheric solar irradiance for band b_n , and $\theta_S(i, j)$ the solar zenith angle at pixel (i, j) . The pipeline enters via the inverse of Equation (3.2) and exits by re-applying it after all radiance-domain degradations; solar irradiance and zenith angle are read from Sentinel-2 tile metadata via the Copernicus Data Space Ecosystem OData API, and no additional solar-irradiance adjustment is needed because Sentinel-2 and Φ -sat-2 spectral bands are close enough in wavelength. Earth–Sun distance is computed from acquisition day-of-year as $d = 1 - 0.01672 \cos(0.9856(\text{day} - 4) \cdot \pi/180)$.

Spatial resampling. Φ -sat-2 has a uniform ground sampling distance of 4.75 m across all bands; Sentinel-2 provides 10 m (bands B2, B3, B4, B8) and 20 m (bands B5, B6, B7). Each band is resampled to the common 4.75 m grid using bicubic interpolation, which limits the resampling artifacts that bilinear interpolation would introduce, though it cannot recover spatial detail the coarser source bands never captured.

Band-to-band misalignment. Push-broom acquisition co-registers spectral bands imperfectly, since detector rows are exposed at slightly different times and platform micro-vibration perturbs the line of sight during each band’s integration window. For the raw L1A product, the displacement of band b_n relative to the previously acquired

output of the pipeline in a new way: we stress-test the base TerraMind encoder across a ladder of degradation severities in order to isolate which physical effect drives the domain gap, and we use it to produce a simulated flood-segmentation dataset, in the absence of any real Φ -sat-2 flood data.

3.3 Domain Gap Measurement

3.3.1 Normalization protocol

Raw Φ -sat-2 digital numbers are heavy-tailed; a square-root transform is applied first to bring the distribution closer to Gaussian, followed by a smooth (soft-saturating) clip toward a domain-specific clip value in place of a hard clip, and finally per-domain z -score standardization using mean and standard deviation statistics computed separately for each sensor domain (real Φ -sat-2, real Sentinel-2, simulated Φ -sat-2). An early baseline was corrupted by a normalization defect that caused weight saturation and an artificially depressed performance estimate; correcting it was a precondition for every result reported in Chapter 4.

The per-domain normalization could itself account for an apparent domain gap due to a difference in raw value scale, so we ran the domain-gap measurement below both with and without it. The two conditions produced similar overall results, and per-domain normalization gave lower cosine similarity between domains in the encoder’s early layers.

3.3.2 Direct and indirect evidence

The domain gap is measured directly, in latent space, using the co-registered triplets: paired cosine similarity between the embeddings of the two sensor views of the same ground, proxy $\mathcal{A}$ -distance (PAD) computed per encoder layer to locate where the gap concentrates, and a cross-domain linear-probe-transfer test in which a probe fitted on one sensor’s embeddings is applied to the other’s embeddings of the same patches. It is measured indirectly through downstream task performance: LULC mIoU of the distilled student relative to a supervised baseline of identical architecture trained directly on the target sensor (Section 3.6), and flood-segmentation IoU across the simulated degradation ladder of Section 3.1.2.

3.4 Domain-Adaptation Strategy

3.4.1 Fine-tuning on degraded data

The first mechanism we attempted was an implicit one: fine-tuning TerraMind’s encoder on the LULC task with degraded, simulated training data, on the premise that the exposure to degraded examples during training would itself induce a more domain-invariant representation. Cross-validation against the real- Φ -sat-2/Sentinel-2 triplets showed that it doesn’t hold. A model fine-tuned this way doesn’t show a lower domain gap to real Φ -sat-2 than a model fine-tuned only on clean Sentinel-2. Standard supervised fine-tuning, even on physics-degraded data, doesn’t translate into a sensor-invariant representation.

$$\mu_d = \frac{\sum_n \sum_{i,j} x_d[i, j, n]}{N \cdot H \cdot W}, \quad \sigma_d^2 = \frac{\sum_n \sum_{i,j} (x_d[i, j, n] - \mu_d)^2}{N \cdot H \cdot W}. \quad (3.13)$$

The running estimates are accumulated separately per domain as well,

$$\bar{\mu}_d^{t+1} = (1 - \alpha) \bar{\mu}_d^t + \alpha \mu_d^t, \quad (\bar{\sigma}_d^{t+1})^2 = (1 - \alpha) (\bar{\sigma}_d^t)^2 + \alpha (\sigma_d^t)^2, \quad (3.14)$$

and at test time the examples of a domain are whitened with that domain's own $\bar{\mu}_d$ and $\bar{\sigma}_d$.

Comparing Equation (3.11) with Equation (3.7), the functional form is unchanged and the entire difference is the subscript d . Every BN layer of a base network $F(\cdot)$ is replaced by a DSBN layer, and each input domain is identified, giving a domain-specific network $F_d(\cdot)$. Only γ_d , β_d , $\bar{\mu}_d$ and $\bar{\sigma}_d$ are duplicated, which for a layer of C channels is four C -vectors per domain, negligible beside the convolutional weights. Everything outside those layers, which is to say every learned filter in the student, stays shared and is trained on both sensors at once. The split is therefore placed exactly at the per-channel scale and shift, the part of the representation that the measurements of Section 3.1.1 identify as the most sensor-dependent, while the filters, which ought to be common, remain so. Chang et al. argue that this is also why the shared weights learn better domain-invariant features: the domain-specific component of the activations can be removed using the statistics and parameters of the domain the input came from.

However, DSBN was introduced for UDA where the target domain carries no labels at all, and the original paper embeds the layer in a two-stage framework that first estimates pseudo-labels for the target domain and then trains on them. We adopt only the layer itself. Our target domain is labelled (Section 3.1.1), both branches are trained under supervision, and no pseudo-labelling stage is involved, so the mechanism is used here in a setting simpler than the one it was designed for.

Figure 3.16 contrasts the two arrangements, with each example activating the branch matching its own domain while the inactive branch is left untouched.

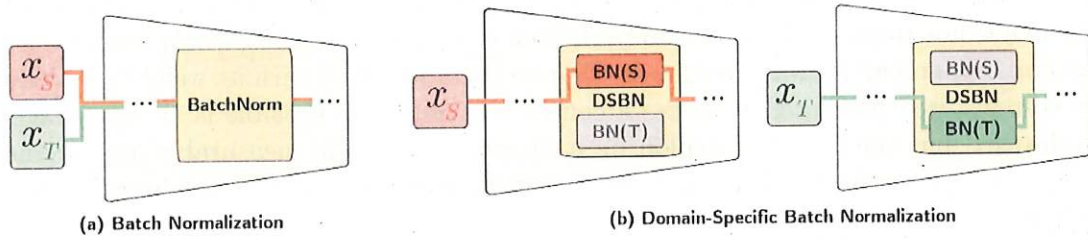


Figure 3.16: Batch normalization compared with domain-specific batch normalization, from Chang et al. [4]. (a) A single BatchNorm layer processes source (x_S) and target (x_T) examples alike, so its statistics describe a mixture of the two domains. (b) A DSBN layer carries one branch per domain, and each input is routed to the branch corresponding to its own domain; the other branch, shown greyed, is inactive for that example. All parameters outside the DSBN layers remain shared across domains. In this thesis the source domain is Sentinel-2 and the target domain is Φ -sat-2, and the domain of every input is known both during training and on board.

Because the deployed sensor on board Φ -sat-2 is always known at inference, selecting the matching normalization branch at deployment time costs nothing operationally, unlike a sensor-agnostic representation, which would need to be invariant without ever being told which sensor produced a given input.

then frozen. The student is a U-Net [38] trained from random initialisation. We chose it for its simple encoder-decoder structure, which is a natural fit for the segmentation task, and for its low operator set requirements, which are compatible with the Myriad 2 VPU, as the Φ satNet baseline demonstrates (Section 2.6).

We considered two placements for a domain-adaptation mechanism: baking the invariance into the frozen teacher and letting the student inherit it passively by mimicking the teacher’s outputs, or giving the student its own adaptation mechanism directly. This thesis adopts the second one, for a reason that is independent of which specific mechanism is used. Two candidate mechanisms were implemented at the student level: domain-specific batch normalization (Section 3.4.2), adopted as the mechanism for the results in Chapter 4, and a paired-contrastive triplet loss (Section 3.4.3), evaluated as an ablation and found not to improve downstream performance further. DSBN also repairs a train/test inconsistency inherent to the two-branch training scheme: the two sensors are forwarded in separate passes, so during training each pass would otherwise be normalized by its own domain’s batch statistics while a single shared set of running averages (a blend of both domains) is used at inference, meaning the encoder would never encounter, at test time, the normalization under which it was optimised. Giving each domain its own running statistics closes this gap as a side effect of the same mechanism.

Figure 3.17 sets out the resulting training scheme. In this configuration, we query the teacher only on Sentinel-2, which is the domain it was pre-trained and fine-tuned on, and where its predictions are most accurate. A single teacher forward pass then supervises both student branches, so the student’s Φ -sat-2 predictions are distilled against a teacher output computed from the Sentinel-2 view of the same ground. This transfer of supervision across sensors is only possible because the two views are co-registered, and it is also what keeps the teacher’s own domain-shift error out of the student. Querying the teacher directly on Φ -sat-2 would distil that error instead of correcting it.

3.5.1 Training objective

Write x_s for the Sentinel-2 view of a patch, x_t for its co-registered Φ -sat-2 view, and y the WorldCover label map. Let $S_d(\cdot)$ denote the student evaluated through the normalization branch of domain d (Section 3.4.2), so that $S_S(x_s)$ and $S_T(x_t)$ are the two forward passes of Figure 3.17, and let $T(\cdot)$ denote the frozen teacher. Every term acts on dense logit maps after bilinear sampling to label resolution. The complete objective is therefore:

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{task}}^T \mathcal{L}_{\text{task}}(S_T(x_t), y) + \lambda_{\text{task}}^S \mathcal{L}_{\text{task}}(S_S(x_s), y) \\ & + \lambda_{\text{KD}} \left[\mathcal{L}_{\text{KD}}(S_T(x_t), T(x_s)) + \mathcal{L}_{\text{KD}}(S_S(x_s), T(x_s)) \right] \\ & + \lambda_{\text{inst}} \mathcal{L}_{\text{inst}} + \lambda_{\text{pix}} \mathcal{L}_{\text{pix}}, \end{aligned} \quad (3.15)$$

and Table 3.1 gives the value of every coefficient.

Task term. $\mathcal{L}_{\text{task}}$ is the focal loss of Lin et al. [39], averaged over the pixels $\mathcal{V}$ that carry a label,

$$\mathcal{L}_{\text{task}}(z, y) = \frac{1}{|\mathcal{V}|} \sum_{(i,j) \in \mathcal{V}} \alpha_{y_{ij}} (1 - p_{ij,y_{ij}})^\gamma (-\log p_{ij,y_{ij}}), \quad p_{ij} = \text{softmax}(z_{ij}), \quad (3.16)$$

with focusing parameter $\gamma = 2$ and per-class weights $\alpha_c \propto f_c^{-1/2}$ derived from the measured pixel frequencies f_c of Section 3.1.1 rescaled to unit mean.

in Chapter 4, so Equation (3.15) reduces there to its first two lines; they are stated here because Section 4.4 reports their ablation.

Symbol	Term	Value
λ_{task}^T	Task loss, Φ -sat-2 branch	1.0
λ_{task}^S	Task loss, Sentinel-2 branch	1.0
λ_{KD}	Distillation, applied to both branches	1.0
λ_{inst}	Instance-level cross-sensor contrast	0
λ_{pix}	Class-conditional pixel contrast	0
γ	Focal focusing parameter	2.0
α_c	Class weights, rescaled to unit mean	$\propto f_c^{-1/2}$
τ	Distillation temperature	4.0

Table 3.1: Coefficients of the training objective, Equation (3.15). The two contrastive weights are zero throughout Chapter 4; the values of γ , α_c and τ are shared by the supervised baseline.

The distilled student is therefore domain-adaptive instead of domain-invariant. The models share the same convolutional weights for both sensors, but not normalization statistics. The correct branch must be selected at inference according to the input source. This is a weaker property than a single sensor-agnostic representation, but it is enough since the sensor domain is known at inference time.

Chapter 4

Results and Discussion

This chapter reports the experiments that evaluate the three research questions of Section 1.3. All the results below use the single configuration set out in Section 4.1 which is a distillation with domain-specific batch normalization, and the comparison of interest throughout is against a supervised baseline of identical architecture trained without any distillation.

4.1 Experimental Configuration

The teacher is TerraMind [11] at the base scale, fine-tuned on the Sentinel-2 branch of the triplet dataset (Section 3.1.1) for the eleven-class WorldCover LULC task [37], and frozen thereafter. The student is a U-Net [38] with 9.85 M parameters, trained from a random initialisation, since no pre-trained weights are available for this backbone. We return to this constraint in Section 4.7.2. Both models consume the seven-band Blue-NIR stack under the per-domain normalization protocol of Section 3.3.1.

Two elements of this configuration need to be stated explicitly, since we arrived at both of them empirically and both of them change the results.

Domain-specific batch normalization. The student carries a separate set of batch-normalization [30] statistics and affine parameters for each sensor, with all convolutional weights shared, following Chang et al. [4]. Beyond its role as a domain-adaptation mechanism, this also corrects a train/test inconsistency that is inherent to the two-branch training scheme. The two sensors are forwarded in separate passes, so during the training each pass is normalized by the batch statistics of its own domain, while a single shared set of running averages, which is a blend of the two domains, is what would be used at inference. The encoder would therefore never encounter at test time the normalization under which it was optimised.

Class-imbalanced task loss. The WorldCover class prior on this dataset is severely imbalanced: the three rarest classes (snow/ice, mangroves, moss/lichen) together account for 1.2% of labelled pixels but, under a macro-averaged mIoU, for 27% of the metric. Under unweighted cross-entropy the student never predicted these classes at all, scoring exactly zero IoU on them in training as well as validation. All the runs reported here therefore use a focal loss [39] with $\gamma = 2$, and an inverse-square-root class weighting derived from the measured pixel frequencies, which is a milder variant of the effective-number scheme of Cui et al. [40].

The teacher's representation does not transfer across the sensor pair. The paired cosine similarity between the two views of the same ground is 0.098, so the two sensors are close to orthogonal in its embedding space, and a linear probe fitted on its Sentinel-2 features retains only 33% of its accuracy once it is applied to the Φ -sat-2 features ($0.754 \rightarrow 0.250$). Figure 4.1 shows this directly: grey lines join the two sensor views of each patch, and in the teacher's projection they stretch across the entire embedding (median link length 6.15 UMAP [41] units).

PAD between the two domains is 1.90–1.98 at every transformer block on a scale of 2. A gap that was semantic in origin would be expected to emerge in the deeper and more abstract layers, whereas one that is present from the first block points instead to low-level radiometry. This is consistent with the spectral analysis of Section 3.3 which found that the two sensors' digital numbers differ largely and that their per-band shapes differ heavily. This anal is also the reverse of the resolution-driven account emphasised in related on-orbit work [15], since the 4.75 m ground sampling distance of Φ -sat-2 is finer than Sentinel-2's.

Same ground, two sensors — grey lines join the paired patches

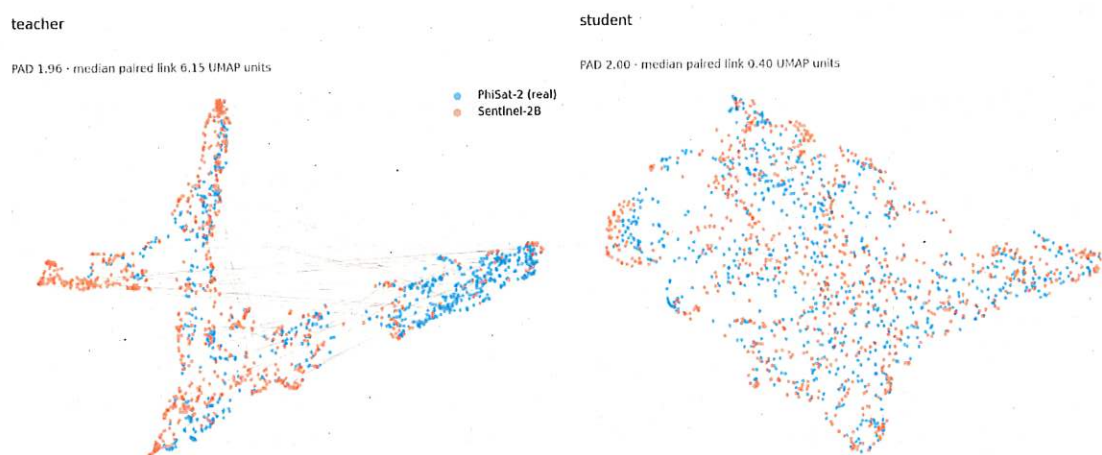


Figure 4.1: UMAP projection of pooled embeddings for 800 co-registered test patches, teacher (left) and distilled student (right), each projected independently. Grey lines join the two sensor views of the same patch. In the teacher's space the paired views are far apart (median link 6.15 units); in the student's they are effectively coincident (0.40 units).

A raw per-layer cosine-similarity check on the smaller TerraMind Tiny encoder, fine-tuned separately on real Φ -sat-2 data as part of the earlier protocol described in Appendix A, is consistent with this picture (Figure 4.2). Similarity between paired Sentinel-2B and real- Φ -sat-2 embeddings is near zero at the first block and rises through the middle of the network, mirroring the low-level-radiometry reading of the per-layer PAD result above. This is a supplementary observation, on a different model scale and with a different measure than the analysis above (a raw cosine similarity in place of PAD), and we offer it as a consistency check rather than as a replication.

Figure 4.3 shows what the four severities of Table 4.2 look like on a representative Sen1Floods scene, with each factor varied in isolation.

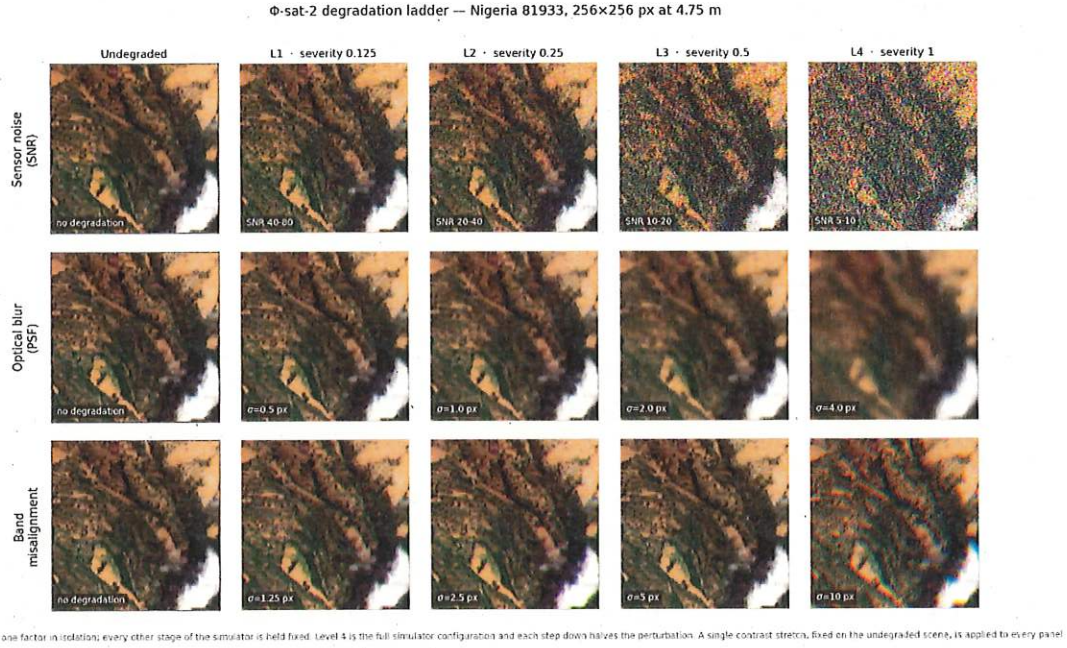


Figure 4.3: A degradation ladder sample from the Sen1Floods benchmark (scene Nigeria 81933, 256×256 px at 4.75 m resolution). Each row varies one factor in isolation across the four severities of Table 4.2, from clean (left) to full severity L4 (right), while every other stage of the simulator is held fixed.

We want to measure the domain gap by isolating the effect of each degradation factor. Each model is trained once on the clean data, and the resulting checkpoint is then evaluated on the test split of every variant, so the degradation only occurs at inference. Three encoder configurations are compared: the distilled student with its encoder frozen and its decoder trained (5.14 M of 9.86 M parameters trainable), TerraMind base under the same frozen-encoder protocol (7.92 M of 94.25 M), and TerraMind base fine-tuned end to end. Each configuration is run for three seeds using the same model checkpoint across all test splits. Table 4.3 reports the results.

On clean data the distilled student is the strongest of the three, at 0.814 IoU against 0.773 for fine-tuned TerraMind and 0.739 for frozen TerraMind. That the smaller student outperforms the foundation model on this clean task can be attributed to its stronger local representation, whereas TerraMind is a general-purpose encoder given only a task head and a fine-tuning pass. The comparison of interest here is the response to degradation.

Band misalignment is the only factor that degrades every configuration significantly, monotonically, and by a similar amount. Its full-severity cost varies only between -0.109 and -0.140 IoU across models that differ in trainable parameters and by architecture family, and it is the only factor resolved at the mildest severity in all three columns, where a 1.25-pixel shift already costs a small but reproducible amount. Its seed-to-seed deviation is also the smallest in the table. A perturbation whose cost barely changes with the model absorbing it is a damage to the input signal, rather than a weakness of any particular representation. Band-to-band shifts corrupt the cross-band relationships on which a water/land decision rests, and no amount of encoder capacity recovers information that

found in related on-orbit work [15].

Factor at L4	Trained clean, tested degraded	Trained, tested degraded	Recovered
Sensor noise (SNR)	-0.455	-0.072	84%
Band misalignment	-0.150	-0.051	66%
Optical blur (PSF)	-0.128	+0.020	all of it

Table 4.4: Degradation seen at inference only, against degradation present during training as well, for the distilled student at full severity.

Repeating the sweep with each model trained on the same degraded condition it is evaluated on, instead of on clean data, recovers most of the loss (Table 4.4). 84% of the noise penalty and 66% of the misalignment penalty disappear, and the blur stops being a penalty, since it scores marginally above the clean baseline, like a mild augmentation. The gap measured in Table 4.3 is therefore largely an adaptation gap rather than an information-loss gap, at least for noise and blur. Where the noise characteristics of the operating sensor are known in advance, simulating them during the training recovers most of the achievable performance, without any domain-adaptation machinery. Misalignment is again the exception: a third of its cost survives training on the degraded condition, which is consistent with reading it as a factor that degrades the cross-band information.

Three limitations bound these results. We vary the factors one at a time, whereas a real sensor pair presents all of them at once, and possibly interacting, so the ladder identifies which effects matter in isolation, and not how they compose. The degradations are simulated, so the conclusion concerns the simulator’s model of the instrument difference, and the absence of real labelled Φ -sat-2 flood data (Section 3.1.2) that makes it necessary. Finally, the supervised U-Net baseline is not included in Table 4.3, so these results characterise the distilled and foundation-model encoders but do not establish whether distillation itself changed robustness relative to training on the target sensor directly.

4.4 Sensor Adaptivity in the Distilled Student

The distilled student demonstrates the opposite behavior across these measurements. The paired cosine similarity rises from 0.098 to 0.948, and the probe transfer from 33% to 97% ($0.663 \rightarrow 0.646$), which means that a classifier fitted on the student’s Sentinel-2 features is very nearly as accurate on its Φ -sat-2 features. In Figure 4.1 the paired links collapse to 0.40 UMAP units. The downstream consequence is visible in the right panel of Figure 4.6. The accuracy of the student on the two sensors converges on parity: on the test split it scores 0.4481 on Φ -sat-2 against 0.4640 on Sentinel-2, which is a residual difference of -0.016 mIoU. Table 4.5 collects the three measures side by side.

Work in progress. To write: a remark on the one row that cuts against the student. The teacher is the stronger encoder within Sentinel-2 (0.754 against 0.663 same-domain probe accuracy), and its advantage only disappears once the probe is asked to cross the sensor gap. This can be framed either as a limitation of the student, or as the precise statement of what the distillation bought.

One measurement in this set behaves counter-intuitively, and it illustrates a real trap in evaluating this class of method. The raw PAD is maximal for the student (2.000,

What PAD measures: separability along the domain-probe axis

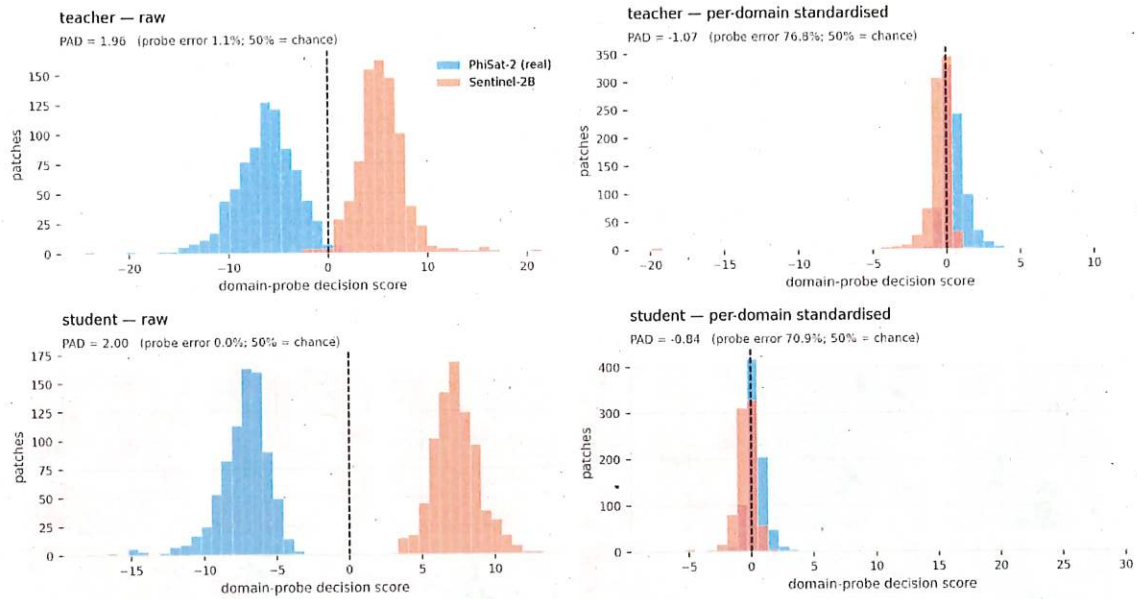


Figure 4.4: Distribution of the domain probe's out-of-fold decision score, the quantity from which PAD is computed. Separated modes imply a probe that never errs (PAD $\rightarrow$ 2); overlapping modes imply chance (PAD $\rightarrow$ 0). After standardising each domain on its own statistics (right column) the separability disappears for both models, showing that all of it was per-channel mean and scale, not representational structure.

4.5 Distillation Against a Supervised Baseline

Sensor adaptivity is only useful if it improves accuracy over a simpler method. The control for this is a student of identical architecture, trained directly on labelled Φ -sat-2 with no teacher, no second sensor and no distillation. Some parameters are held constant for fairness: the split, the normalization, the augmentation, the task loss, the optimiser, the schedule and the early-stopping criterion (Table 4.1).

Labels	Method	Val. mIoU	Test mIoU	Cost (h)
10,000	KD + DSBN	0.4607	0.4481	13.3
10,000	Supervised baseline	0.3720	0.3739	4.0
1,000	KD + DSBN	0.2679	0.2635	2.2
1,000	Supervised baseline	0.2916	0.2707	0.6
TerraMind teacher (Sentinel-2, reference)		—	0.6351	—

Table 4.6: Main results on the LULC task. Validation mIoU is the best epoch, on the fixed 1,000-patch validation split shared by every run. Test mIoU is on the full 25,323-patch test split. Cost is total wall-clock for the run.

At a budget of 10,000 labelled patches the distilled student has 0.4607 best validation mIoU against the baseline's 0.3720, an improvement of +0.0887 and 0.4481 against 0.3739 on the test split (Table 4.6). Figure 4.6 shows the shape of that margin: the baseline is

ahead during the first twenty-five epochs, then the two curves cross but the distilled run keeps climbing after the baseline has flattened and stalled due to early stopping.

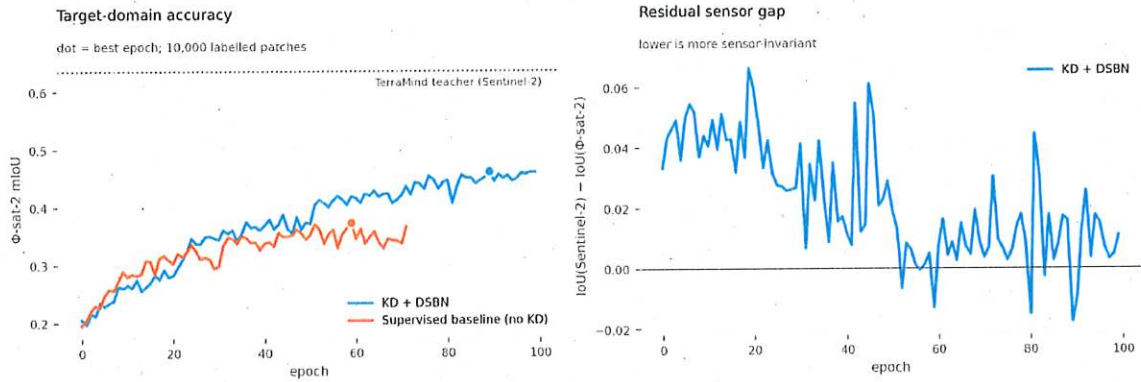


Figure 4.6: Left: Φ -sat-2 validation mIoU over training at 10,000 labelled patches, with the best epoch marked. Dotted line marks teacher’s accuracy on Sentinel-2. Right: the residual accuracy difference between the two sensors for the distilled student, which crosses zero around epoch 55 and oscillates.

The distilled run’s best epoch is the 89th of the 101 it ran, and it exhausted the full budget without triggering the early stopping, so the result should be read as a lower bound on what this configuration reaches. The baseline, by contrast, early-stopped at epoch 73.

The per-class results (Figure 4.7) show that the gain is broad. Distillation is ahead on ten of the eleven classes, most markedly on moss/lichen (+0.148), snow/ice (+0.124), mangroves (+0.120), cropland (+0.112), tree cover (+0.097) and grassland (+0.096). Permanent water is the only exception (−0.013). The frequency-weighted IoU moves in the same direction as the macro average (0.523 against 0.454), so this result is not an artefact of the rare-class weighting introduced in Section 4.1.

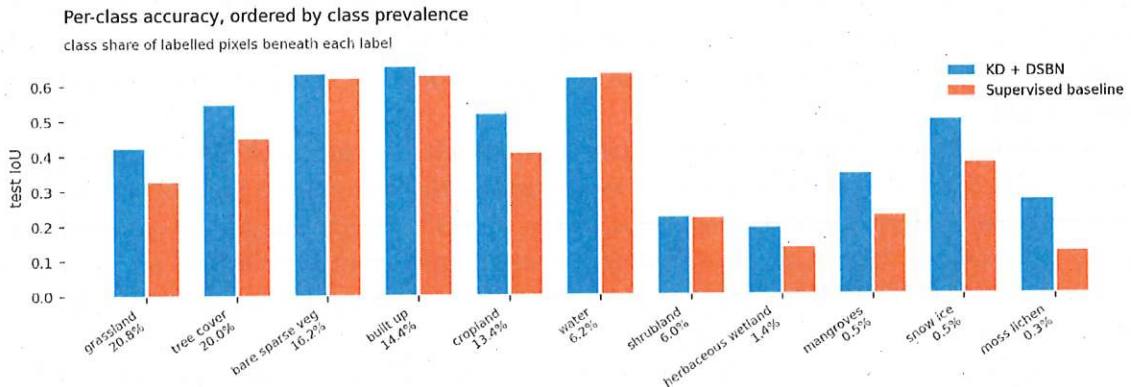


Figure 4.7: Per-class test IoU at 10,000 labelled patches, ordered by class prevalence, with each class’s share of labelled pixels beneath its label. Neither model leaves any class unlearned, which unweighted cross-entropy did for four of them (Section 4.1).

The TerraMind teacher has 0.6351 mIoU on Sentinel-2 while the student reaches 0.4481 on Φ -sat-2. This difference is a compression gap rather than a domain gap,

observations point the same direction. Zero-shot with real Φ -sat-2 on the released land-cover model scores between 0.0008 and 0.0413 mIoU across the ten published checkpoints, against the 0.0376 that a uniform-random predictor scores on the eleven class frequencies. On this task and at this budget it is therefore the architecture, and not the pre-training that accounts for the Φ satNet result.

4.6 Accuracy Drop with Few Labels

KD and **UDA** are usually motivated by a label scarcity in the target domain [26], so we would expect the advantage above to grow as the target labels are removed; the opposite is what we observe. At 1,000 labelled patches the supervised baseline is ahead by every measure: best validation mIoU 0.2916 against 0.2679, test mIoU 0.2707 against 0.2635, and on matched epochs the distilled student leads in only 8 of the 58 epochs both runs share. It also does so at roughly a third of the cost, and this even though the distilled run had the larger budget available to it: it used its full hundred epochs without triggering the early stopping, while the baseline stopped at sixty-seven. Figure 4.8 shows the crossover with both budgets evaluated on the identical fixed 1,000-patch validation split. One reading is that the distilled configuration carries substantial overhead, which is a second forward pass, doubled normalization parameters, and a distillation term pulling toward a teacher, and that below a certain label threshold the cost exceeds the value of the teacher’s signal.

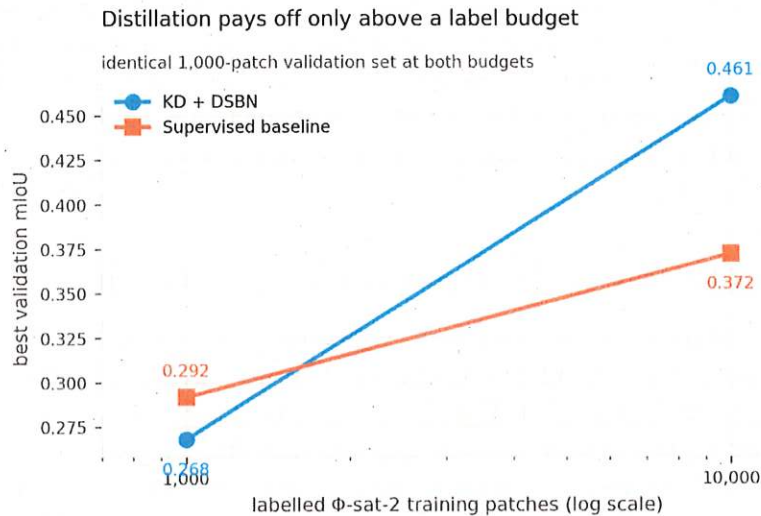


Figure 4.8: Best validation mIoU against labelled training budget. The two methods cross: distillation is worse at 1,000 labelled patches and better at 10,000. Both budgets share the same fixed validation split.

4.7 Discussion

4.7.1 Domain-specific normalization contribution

The improvement brought by the domain-specific normalization is larger than what a pure domain-alignment effect would predict. The accuracy on the source sensor improved by a

lightweight model already achieves adequate performance, what is gained by adapting TerraMind ?

Table 4.7 answers the accuracy half of that question at a budget of 10,000 labels. Φ satNet reaches 0.3316 test mIoU against the supervised U-Net's 0.3739 and the distilled student's 0.4481, with 28 times fewer parameters than either. Per parameter it is far ahead of both; in absolute accuracy it is behind both. Which of those two readings should govern is a matter of deployment design, because under the power and memory budget of an on-board processor it is not obvious that the absolute reading wins.

The matched initialisation result shows that the pre-training contributes almost nothing: fine-tuning from the released land-cover checkpoint beats the identical architecture from a random initialisation by 0.0102 test mIoU. The same checkpoint, run on real Φ -sat-2 without any training, scores at the level of a uniform-random predictor. What Φ satNet contributes in this comparison is its parameter efficiency and not transfer from its mission-specific corpus.

This is not an evidence against the claim in [21], which concerns a different benchmark suite. However, this thesis addresses the same issue: the released Φ satNet weights were pre-trained on simulated Φ -sat-2 imagery, and Section 4.2 shows that a gap of this kind originates in low-level radiometry rather than in semantics, which is the part of a sensor a simulator has to replicate.

Distillation from a GeoFM does not close the gap either which is a significant finding. Handing Φ satNet the same teacher, objective and budget that lift the U-Net by 0.074 mIoU doesn't help here (Section 4.5). This places the limit on capacity rather than on supervision. The choice this poses for an on-board system is therefore not between a well-taught small model and a poorly-taught large one: below some size the teaching stops arriving, and the only way to buy the distilled student's accuracy is to carry its parameters. Each configuration here is a single seed, so the margins of order 0.01 separating the Φ satNet variants should be read as the absence of a large effect rather than as measurements of small ones (Section 4.7.2).

4.7.5 Practical recommendations

These findings suggest three practical recommendations. First, missions intended to host on-board foundation models should prioritise collecting co-registered calibration data early in their operational life. The alignment with some existing acquisition from Sentinel-2 made the measurements of this chapter possible. Second, when the deployed sensor is known at inference, domain-specific batch normalization is a valuable mechanism to bridge the domain gap as it only adds a few thousand additional parameters to the model. Third, the label budget should be measured before adopting a distillation pipeline, since below the crossover of Section 4.6 the simpler supervised baseline was more accurate and faster to train.

at a much smaller one, so the advantage reverses between the two budgets, although the small-budget end of that reversal falls within the run-to-run variance this thesis is able to resolve. Treating the domain-gap problem and the compute-constraint problem as one single combined training objective, and not as two sequential and independent fixes, is the central methodological position of this thesis, with the important qualification that whether this combination is worth adopting depends on how many target-domain labels are available.

This work is limited to a single mission and a single sensor pair, and to one specific pre-trained **GeoFM**. Whether the spectral-response-driven account of the domain gap generalizes to other on-board missions, and whether TerraMind’s broad pre-training is in fact more label-efficient than a mission-specific model trained from scratch, are both open questions that this thesis frames clearly but does not close. Further comparisons against Φ satNet are the immediate next step toward closing the second one. An extension of this work would then be to repeat the analysis for other sensor pairs and other missions, or to attempt to train a custom foundation model instead of adapting an existing one, in order to test whether the findings here are specific to TerraMind, or general to the class of large and diversely pre-trained **EO** models it represents.

Appendix A

Real-Data LULC Baseline

The results in Chapter 4 are built around a source-vs-target evaluation protocol: a teacher fine-tuned only on Sentinel-2, a student distilled and adapted to Φ -sat-2, and comparisons of both against real Φ -sat-2 ground truth. An earlier protocol fine-tuned TerraMind directly on real Φ -sat-2 LULC data with no domain-adaptation mechanism and no second sensor involved. We do not incorporate this evaluation into the main results, because a model trained and evaluated on the same sensor domain doesn't say whether a domain gap exists or has been closed, which is the research questions stated in Section 1.3

A.1 Protocol

Two TerraMind variants, Tiny and Base, were fully fine-tuned on the real- Φ -sat-2 branch of the triplet dataset (Section 3.1.1) for the eleven-class WorldCover LULC task, following the per-domain normalization protocol of Section 3.3.1. Both were trained and evaluated on real Φ -sat-2 data only. We also fine-tuned and evaluated both variants on the real Sentinel-2 branch of the triplet dataset, to establish a baseline for the source domain.

A.2 Result

TerraMind fine-tuned this way achieves mIoU 0.47 (Tiny) and 0.51 (Base) on the held-out real test split. This establishes that the LULC task is learnable on real Φ -sat-2 data, and that it was not a bottleneck for the main results. We also observe that the GeoFM fine-tuned on the real Sentinel-2 data achieves mIoU 0.56 (Tiny) and 0.63 (Base) on the Sentinel-2 test split, which is a drop of 0.09 and 0.12 from the Φ -sat-2 test split, respectively.

Measure	TerraMind Tiny	TerraMind Base
Test mIoU Φ -sat-2	0.47	0.51
Test mIoU Sentinel-2	0.56	0.63

Table A.1: Real-data LULC baseline results on the test split of the triplet dataset. Models are fine-tuned on their respective sensor domain only, and evaluated on the same domain.

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